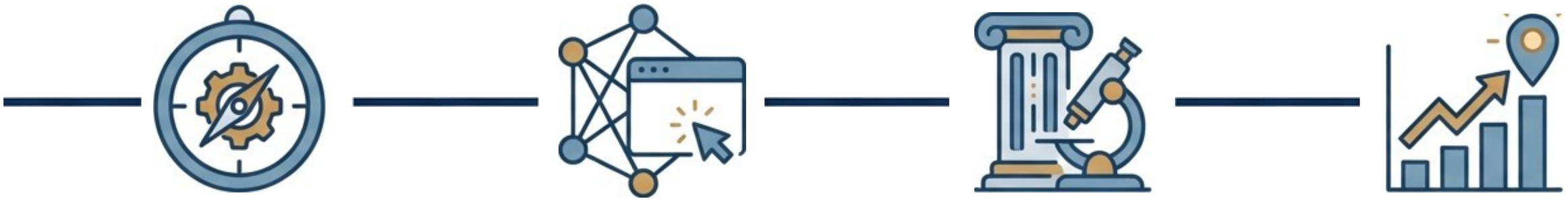
Artificial Intelligence & Applied Research at CSE/HCMUT

From Core Architectures to Real-World impact

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Outlines



1. Introduction to AI

What is it? How does it work? What are the milestones?

2. Potential Applications

Where is the technology being utilized?

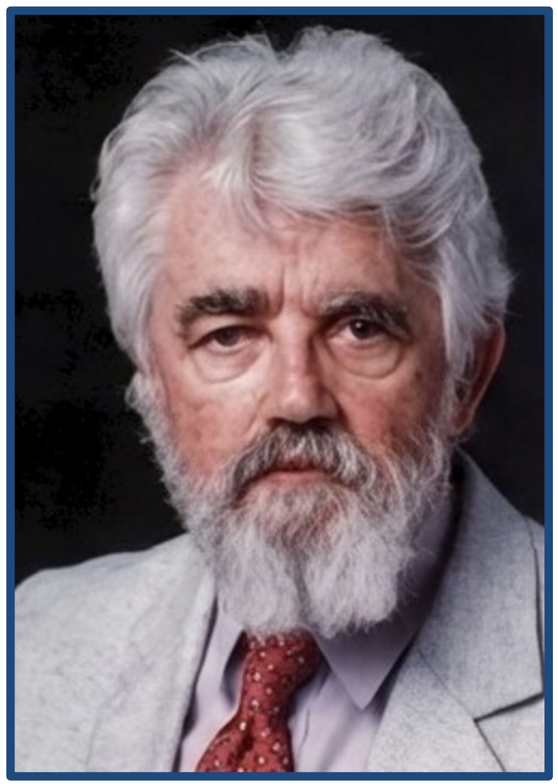
3. Research at CSE/HCMUT

Advancing the frontier at the university.

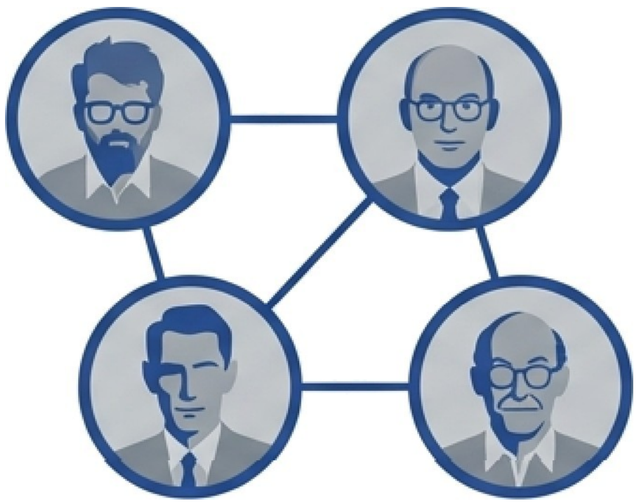
4. Trends in Vietnam

Local development and application vectors.

The Genesis of Artificial Intelligence



"A field of research aimed at equipping machines with human capabilities."



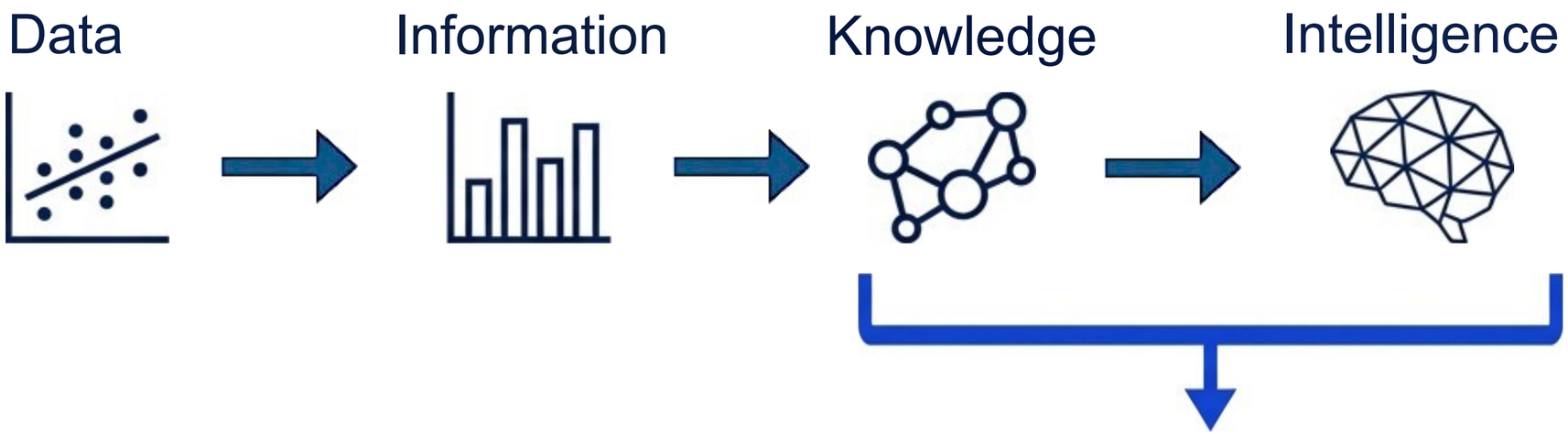
The Dartmouth Summer Research Project on Artificial Intelligence (1956)

John McCarthy | Allen Newell | Claude Shannon | Marvin Minsky

The Quest for Machine Intelligence

Dartmouth 1956

The origin point: defining AI as the science of giving machines human capabilities (seeing, hearing, understanding).



Reasoning based on knowledge remains the weakest capability and the primary focus of the current research community.

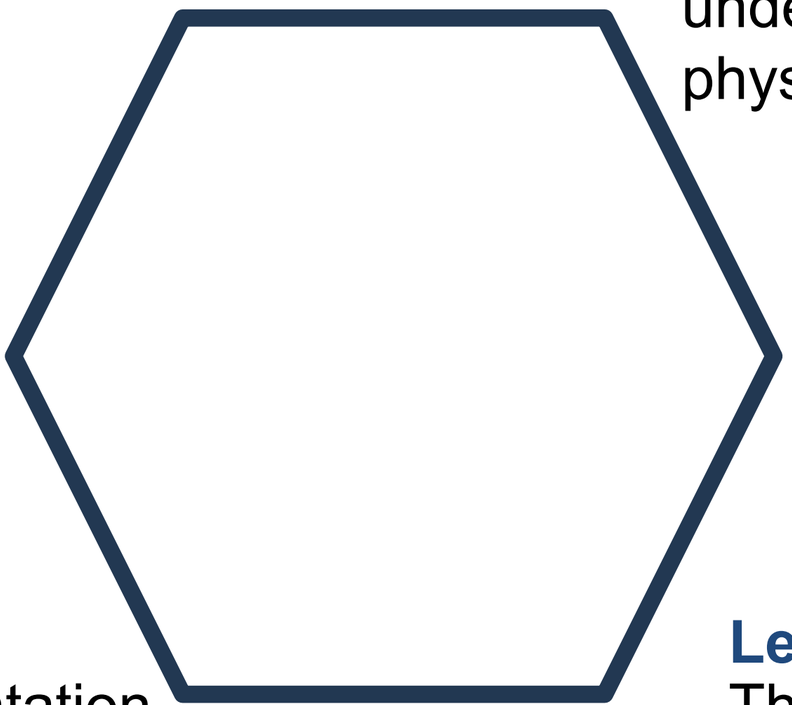
The Anatomy of Intelligence

What special capabilities do humans actually have?

The Human Blueprint

Communication
Voice recognition, speaking

Vision
Seeing and understanding the physical world.



Language
Understanding documents, summarization, writing

Action
Robotics, movement, physical coordination, and driving.

Reasoning
Knowledge representation and deduction. The ultimate, most complex frontier.

Learning
The foundational mechanism, lifelong skill acquisition.

Human Capability in the Era of AI

THE PILLARS:
SENSE, SYMBOL, AND ACTION

Sensory Perception
The ability to process images, recognize speech, and understand video input.

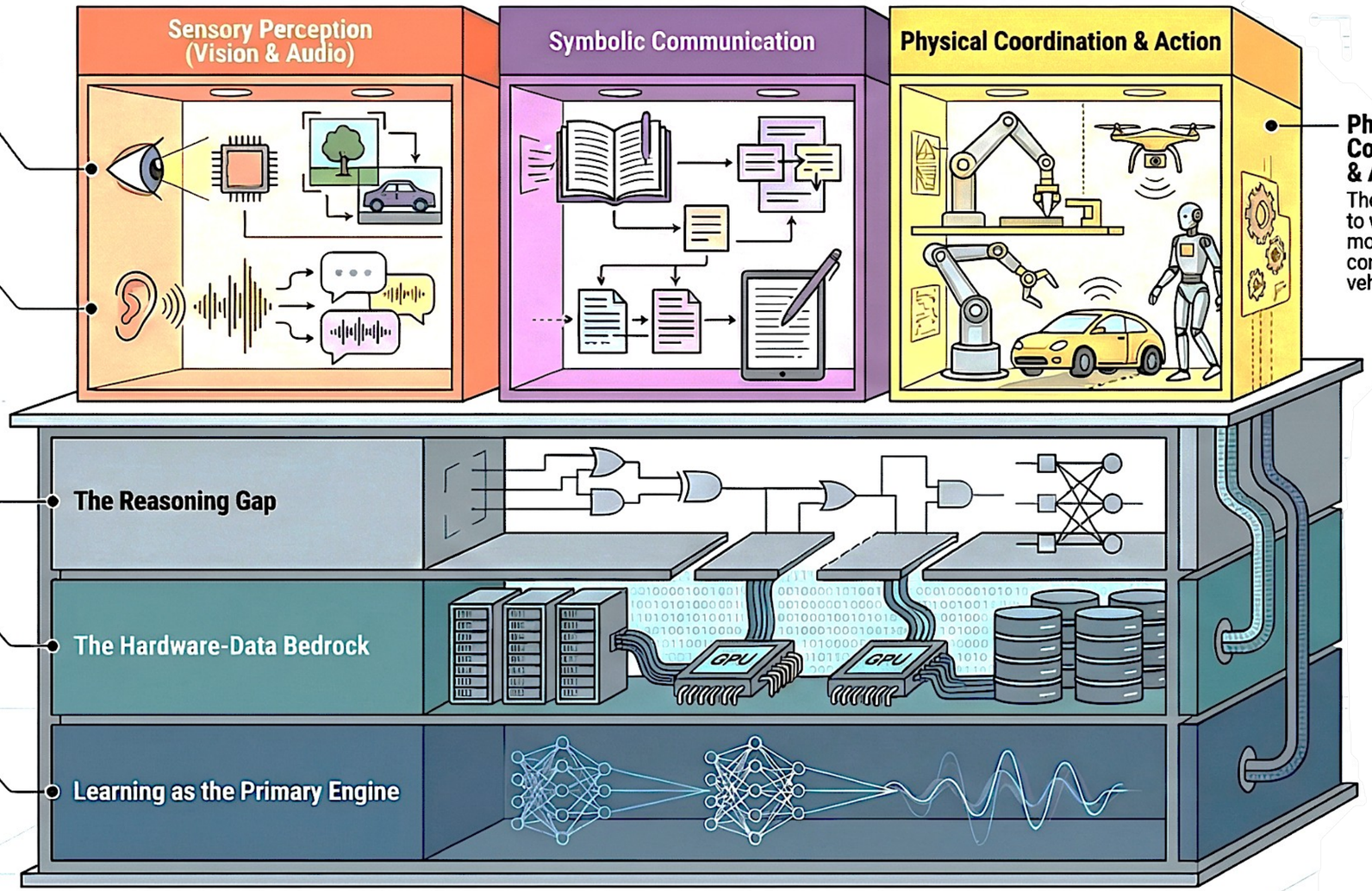
Symbolic Communication
Proficiency in reading, writing, and summarization to facilitate human-like understanding.

THE FOUNDATION:
ENGINE & INFRASTRUCTURE

Symbolic reasoning and inference remain the most challenging and least-developed areas of intelligence.

The Hardware-Data Bedrock
High-performance GPUs and massive training datasets provide the necessary power for intelligence models.

Learning as the Primary Engine
Continuous learning is the fundamental technique required to build all other cognitive skills.



Physical Coordination & Action
The complex ability to walk, coordinate movements, and control autonomous vehicles or drones.

The Philosophy of Learning



The Human Context

- The Vietnam Ethos: “Học, học nữa, học mãi” (Learn, learn more, learn forever).
- The US ABET Engineering Standard: “Lifelong learning.”



The Machine Context

Continuous data ingestion, algorithmic refinement, and perpetual model training to iteratively achieve and scale intelligence.

The Computational Engine: AI as a Mathematical Function

$$Y = F(x)$$

Input (x)
Massive volumes of high-quality, multi-modal data.


Images


Text


Signals

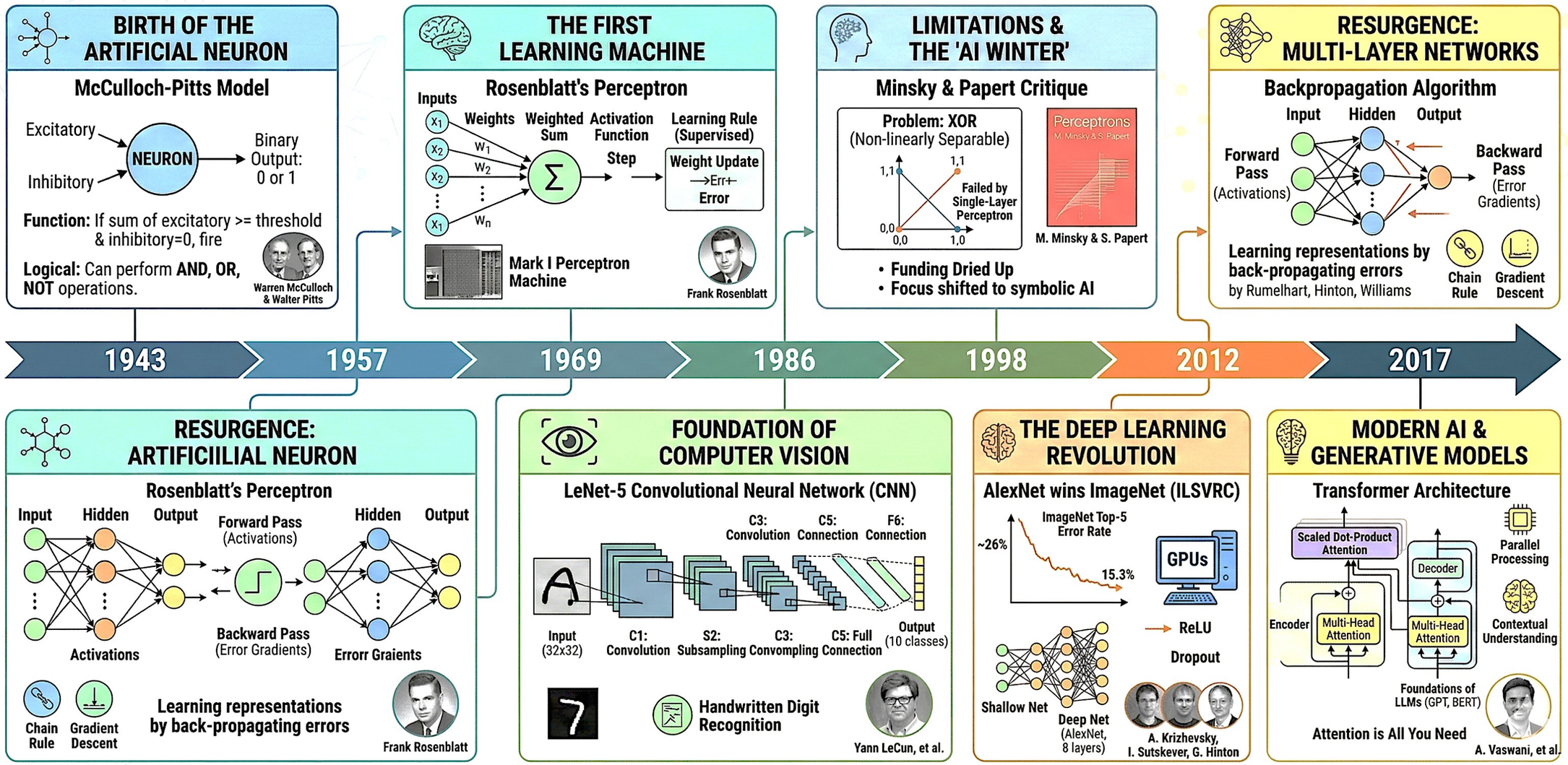
Function (F)
The AI computational model processing the data.

Output (Y)
Generated data or intelligent actions (human-like robotics, autonomous navigation).



Requirement: Massive computational hardware (GPUs) to execute.

The Evolution of Neural Network



2012: The Vision Breakthrough

The Milestone

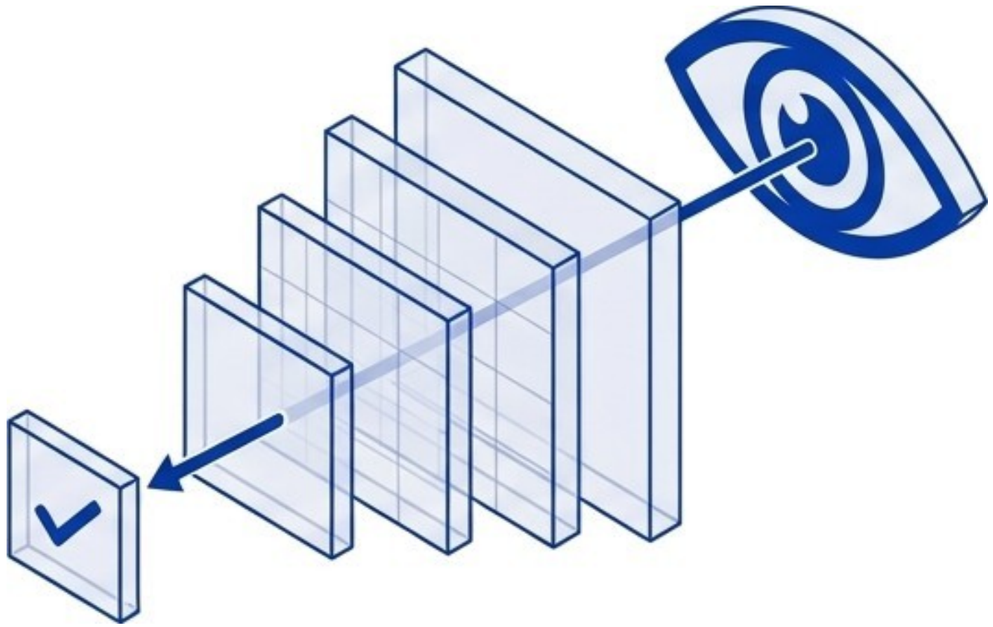
AlexNet & Convolutional Neural Networks (CNNs)

The Shift

Unified computing techniques utilizing 1D, 2D, and 3D convolutions for signal processing.

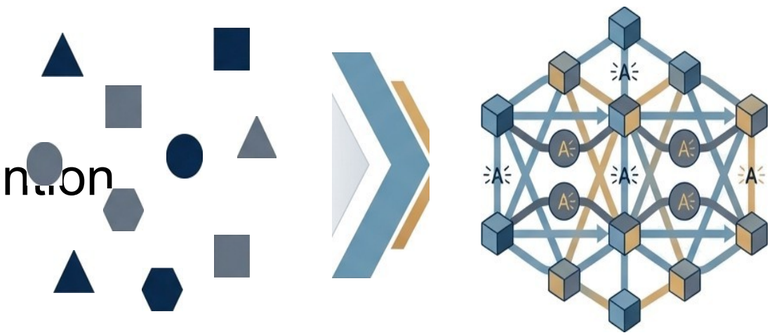
The result

Machines rapidly began mastering human-level visual understanding

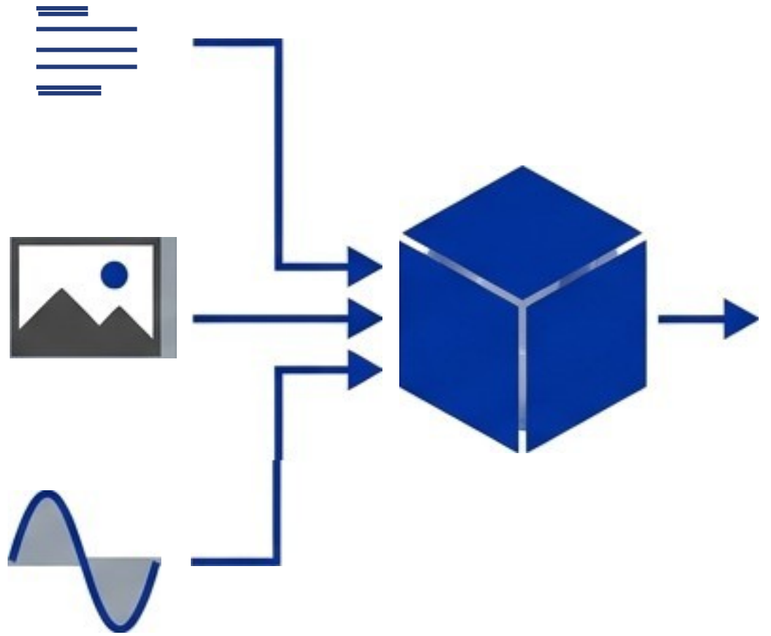


2017: The Architecture Revolution

The Milestone
Transformers & The Attention Mechanism.



The Catalyst
Emphasized self-supervised learning, permanently removing the slow bottleneck of human data labelling.



Unified Structure
A single architecture capable of processing diverse data modalities (text, images, time-series) identically.

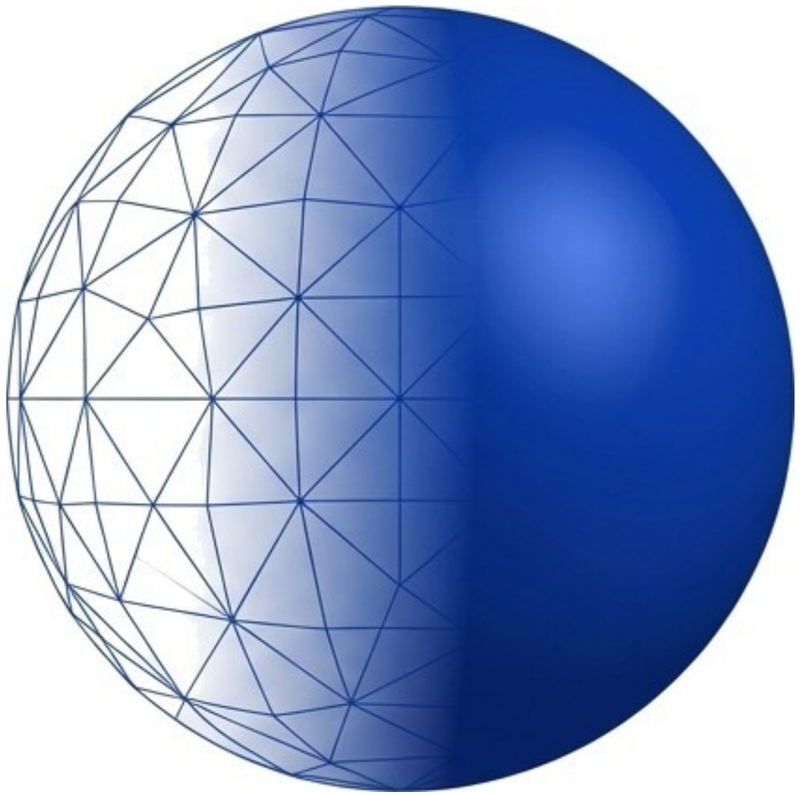
Nov 30th 2022: The Generative Leap

The Milestone

ChatGPT is introduced to the public.

The Achievement

The first AI system capable of utilizing language seamlessly, marking the explosive birth of mainstream Generative AI.



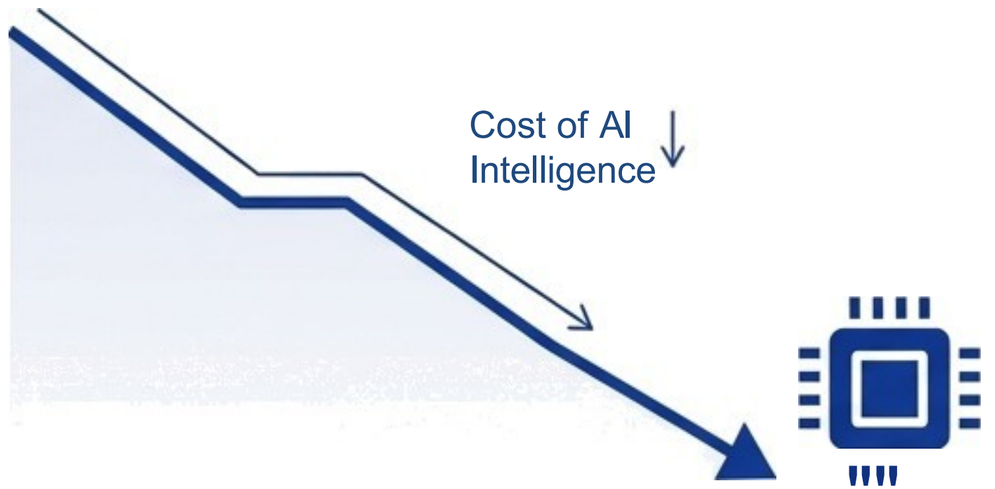
Jan 20th 2025: The Era of Accessibility

The Milestone

DeepSeek-R1 introduced.

The Trend

The definitive rise of high-performance, low-cost AI models. Intelligence is becoming a cheap, abundant utility.



The Reality of Generative AI

Definition

AI systems capable of creating high-quality, original content including text, images, and audio.

Language & Vision:



Scoring 8-9/10 on human capability scale

Action:



Excelling on human-like robotic and autonomous vehicles

Reasoning:



The final remaining frontier for research community

The Global Economic Impact

+$2.6 - $4.4 Trillion

Projected annual boost to the global economy

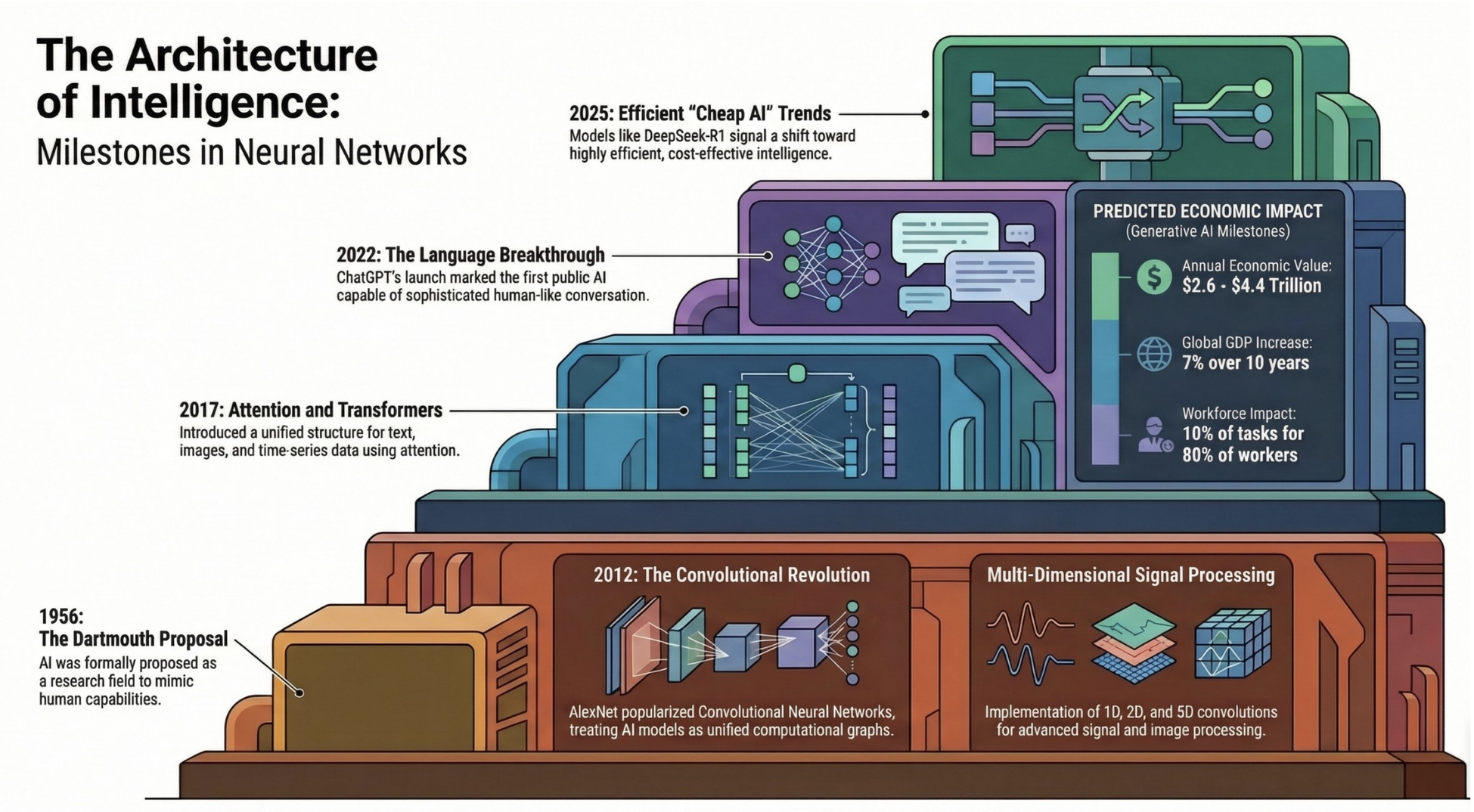
+7%

Expected increase in Global GDP over the next decade

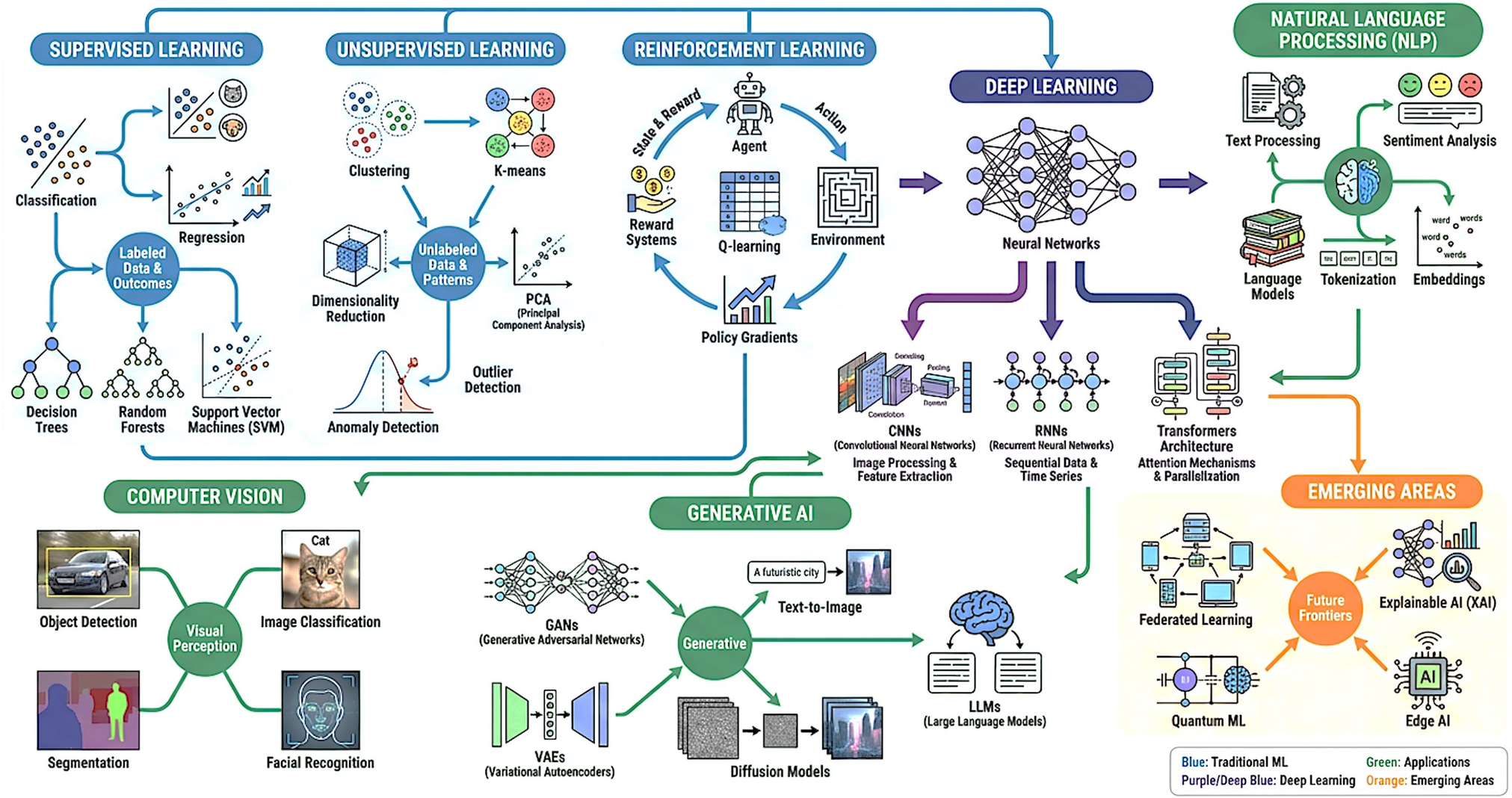
10%

The amount of daily work tasks that will be impacted for 80% of the global workforce

Recap


The Architecture of Intelligence:
Milestones in Neural Networks
2025: Efficient "Cheap AI" Trends
Models like DeepSeek-R1 signal a shift toward highly efficient, cost-effective intelligence.
2022: The Language Breakthrough
ChatGPT's launch marked the first public AI capable of sophisticated human-like conversation.
2017: Attention and Transformers
Introduced a unified structure for text, images, and time-series data using attention.
1956:
The Dartmouth Proposal
AI was formally proposed as a research field to mimic human capabilities.
PREDICTED ECONOMIC IMPACT
(Generative AI Milestones)
Annual Economic Value:
$2.6 - $4.4 Trillion
Global GDP Increase:
7% over 10 years
Workforce Impact:
10% of tasks for 80% of workers
2012: The Convolutional Revolution
AlexNet popularized Convolutional Neural Networks, treating AI models as unified computational graphs.
Multi-Dimensional Signal Processing
Implementation of 1D, 2D, and 5D convolutions for advanced signal and image processing.

Major topics in AI & ML



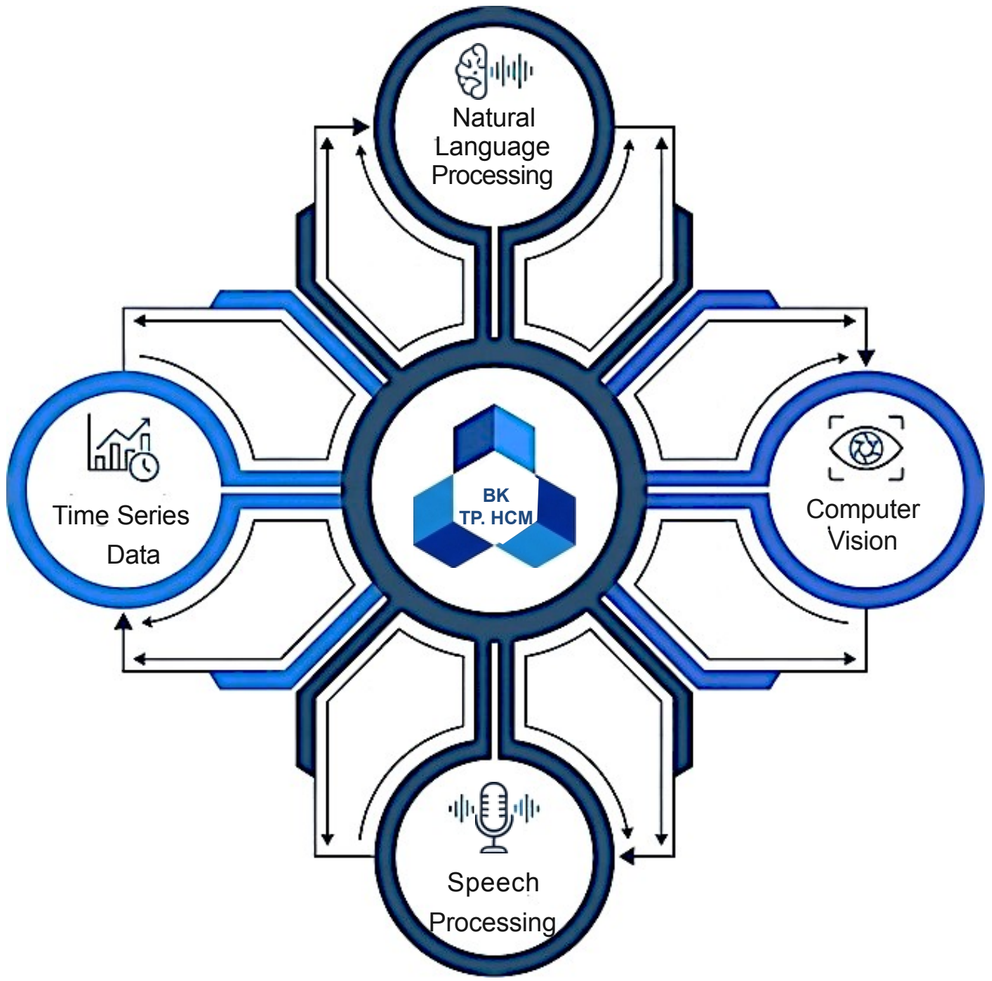
The URA Lab: Unlimited Research Group of AI

Leadership

Directed by Assoc. Prof. Quan Thanh Tho.

Mission

Developing advanced AI solutions for education, healthcare, and industry.



Advancing Natural Language Processing

Language Models

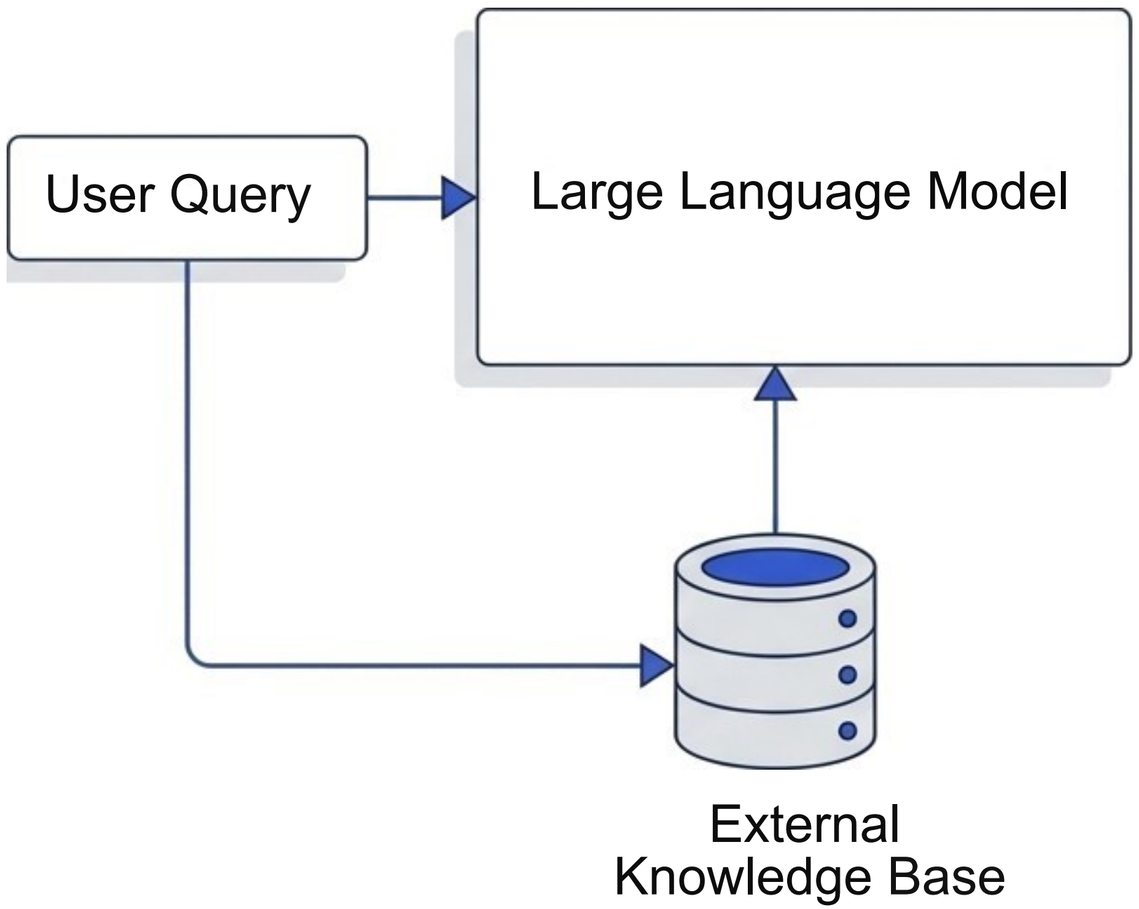
Custom LLM development for Vietnamese and the Ba Na ethnic minority language.

Advanced Systems

Integrating Knowledge Representation and Symbolic Reasoning into robust QA Systems.

Cutting-Edge Techniques

Improving Retrieval-Augmented Generation (RAG) and exploring Multi-Modal Models.



NLP in Action: Institutional & Cultural Impact

HCMUT Chatbot

Pioneering Vietnamese LLM (URA-LLaMa-2.1 8B) deployed for university admissions consulting.



Minority Language Preservation

State-level scientific project delivering automated machine translation and synthetic, realistic text-to-speech for the Ba Na people.



Pushing the Boundary of Computer Vision



Healthcare & Agriculture:
Knee osteoarthritis diagnosis and plant leaf disease classification.



Generative Vision:
Stable Diffusion Models powering virtual try-on systems (showcased to the Prime Minister at VNU's 30th Anniversary).

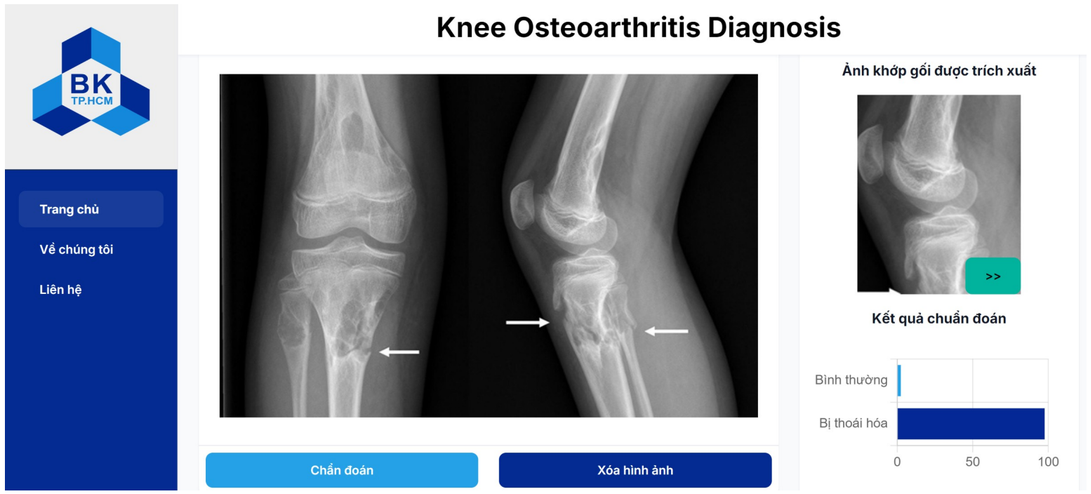


Core Tech:
Visual Question Answering and advanced Energy-Based Models for image classification.

Vision in Action: Explainability & Automation

Explainable Healthcare AI

Doesn't just predict knee osteoarthritis severity, it analyzes X-rays, highlights specific bone abnormalities, and explains the diagnosis.





Smart Retail

Automated warehouse monitoring and inventory classification using real-time object detection.

Beyond the Cloud: Signals, Speech, and Edge AI

Speech & Signals

Denoise tools, Spoken QA systems, and deep learning for Lithium-ion battery capacity prediction. Signal action prediction via EEG/ECG.



RISC-V Edge AI

Running YOLOv5 directly on low-power, edge-computing cameras. Enables on-device object classification, face recognition, and skeletal tracking without relying on massive cloud GPUs.



The Architecture of Impact

Summary
Foundational shiks in AI architecture have transformed computation from pure mathematical abstraction into tangible, real-world tools.

The HCMUT Commitment
Through the URA Lab, CSE/HCMUT is leveraging these global breakthroughs to build localized models, preserve national languages, and pioneer explainable AI for the future of Vietnam.